PROTEIN FOLDING

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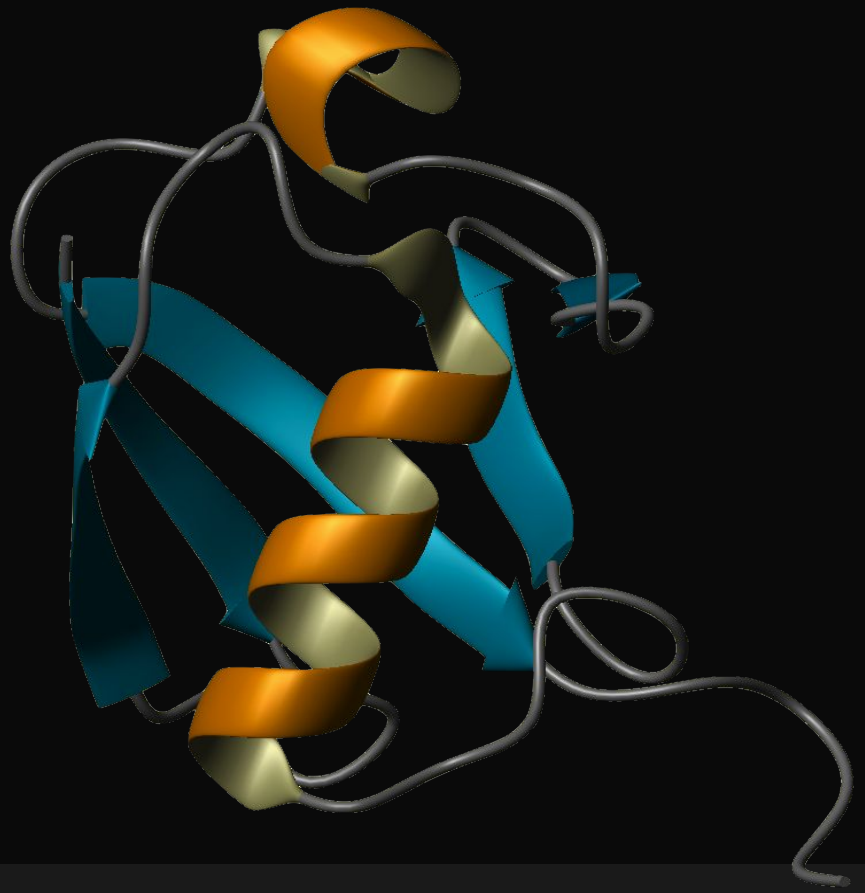


Quantum simulation of a scientifically relevant problem

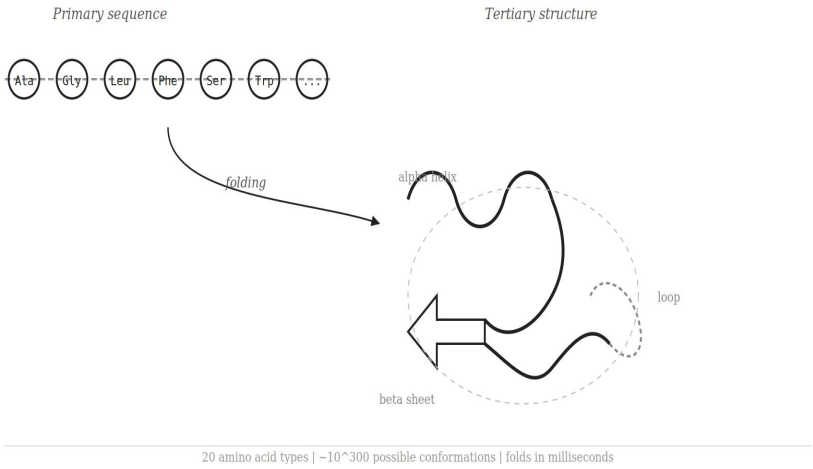
01

What is Protein Folding?

Proteins are the molecular machines of life.

A linear chain of amino acids encodes a unique 3D structure and that structure dictates function. Misfolded proteins drive Alzheimer's, Parkinson's, cystic fibrosis, and prion diseases.

Understanding folding = understanding disease at the molecular level.



"Nature isn't classical, dammit, and if you want to make a simulation of nature, you'd better make it quantum mechanical."

— Richard Feynman, 1981

04

AlphaFold & the 2024 Nobel

WHAT IT DOES

Hassabis & Jumper (DeepMind) won the 2024 Nobel Prize in Chemistry for AlphaFold2.

Deep learning on MSAs + structural templates predicts 3D coordinates from sequence with near-experimental accuracy (GDT > 90 on CASP14).

200M+ structures predicted. Revolutionised structural biology overnight.

WHAT IT DOESN'T DO

AlphaFold predicts structure, it doesn't explain folding.

No folding dynamics or pathway information. No free energy landscape. No quantum effects captured. No conformational ensembles for IDPs. Accuracy degrades for novel folds without homologs.

It's an interpolation engine, not a physics simulator.

02

Why Is Folding Hard?

LEVINTHAL'S PARADOX

10^{300}

possible conformations for a typical protein yet it folds in milliseconds. Exhaustive search is cosmologically impossible.

ENERGY LANDSCAPE

$E(r)$

The free energy surface is rugged, high-dimensional, and riddled with local minima. Classical MD can't reliably find the global minimum.

QUANTUM EFFECTS

$H|\psi\rangle$

Hydrogen bonding, proton tunnelling, and dispersion forces are inherently quantum. Classical force fields approximate — they don't capture the physics.

Protein folding is exponentially hard, fundamentally quantum, and clinically urgent.

03

NP-Hard on a Lattice, Native on an Ising

ISING HAMILTONIAN

$$H = \Sigma_{i\square} \; J_{i\square} \; \sigma_i^z \; \sigma_\square^z \; + \; \Sigma_i \; h_i \; \sigma_i^z$$

Turn encoding → binary variables → Pauli-Z operators
Self-avoidance + overlap → quadratic penalty terms

WHY NP-HARD

Even 2D HP is NP-complete (Berger & Leighton, 1998).

Maximising non-bonded H-H contacts on a self-avoiding walk. Conformations grow exponentially — ubiquitin at 76 residues yields ~10^{47} configurations. Classically intractable.

WHY ISING

Every NP problem reduces to Ising in polynomial time (Lucas, 2014).

The Ising Hamiltonian is the universal language of combinatorial optimisation on quantum hardware. QUBO and Ising are related by z = 2x - 1. Ground state of H_Ising = optimal fold.

Berger & Leighton (1998) J. Comput. Biol. | Perdomo-Ortiz et al. (2012) Sci. Rep. | Robert et al. (2021) npj Quantum Inf. | Lucas (2014) Front. Phys.

05

From Amino Acids to the HP Model

Proteins are chains of amino acids.
There are 20 different types.

Hydrophobic interactions dominate the folding process

We simplify the description of amino acids by classifying them as:

- hydrophobic (H)
- polar (P)

Real sequence of amino acids

M Q I F V K T L T G K T I T L E V E P S D T I E N V K A

HP sequence

H P H H H P P H P P P P H P H P P H P P P H P H P H

We retain the dominant physics in a simplified effective model.

06

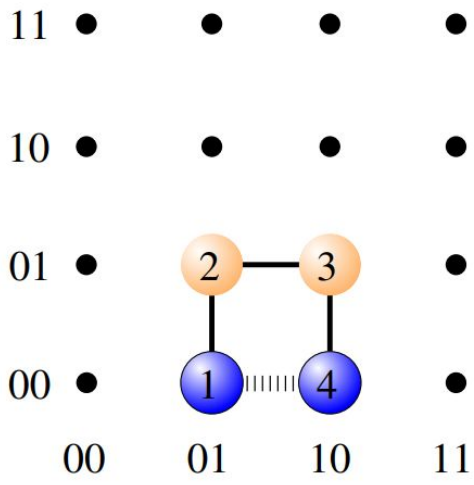
HP Model: Geometry

The protein is placed on a **D-dimensional lattice.**
Each amino acid occupies one lattice site.

We encode the lattice coordinates of each amino acid in binary.
For a 4 amino acid peptide:

$$q = \underbrace{q_{16}q_{15}}_{y_4}\ \underbrace{q_{14}q_{13}}_{x_4}\ \underbrace{q_{12}q_{11}}_{y_3}\ \underbrace{q_{10}q_{9}}_{x_3}\ \underbrace{q_{8}q_{7}}_{y_2}\ \underbrace{q_{6}q_{5}}_{x_2}\ \underbrace{q_{4}q_{3}}_{y_1}\ \underbrace{q_{2}q_{1}}_{x_1},$$

In general, the needed bit per amino acid are $D \log_2 N$



$$q = 0010\ 0110\ 0101\ 0001$$

The structure is encoded as a binary representation of lattice coordinates.

07

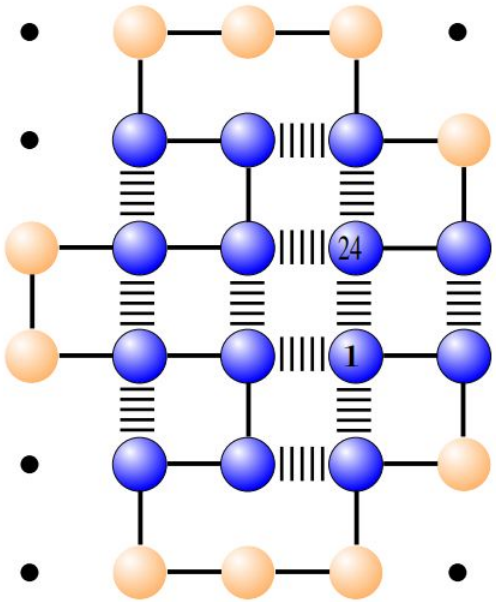
HP Model: Energy

Interactions

Only non-consecutive H–H contacts are energetically favored.
Each contact contributes −1.
Non-consecutive H–P, P–P → no contribution.

Physical consequence

Hydrophobic residues cluster together.
Formation of a compact core.

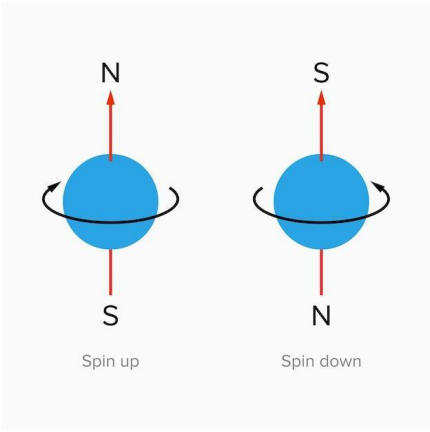


Folding is driven by maximizing hydrophobic contacts.

08

Mapping

Classical	*QM*
$q = q_{16}q_{15}q_{14} \cdots \cdots . q_2 q_1$	$\lvert q\rangle = \lvert q_{16}\rangle \lvert q_{15}\rangle \ldots . \lvert q_2\rangle \lvert q_1\rangle$
q_i	$\widehat{q_i} \equiv \frac{1}{2}\,(I - \hat{\sigma}_i^z)$
e.g. 0101	$\lvert 0101\rangle \equiv \lvert\downarrow\uparrow\downarrow\uparrow\rangle$



09

Adiabatic Quantum Computation

$$\left|\psi_g(0)\right\rangle = \frac{1}{\sqrt{2^n}} \sum_{q_i \in \{0,1\}} |q_n\rangle|q_{n-1}\rangle \dots |q_2\rangle|q_1\rangle$$

$$\widehat{H}(0) = \sum_{i=1}^{n} \frac{1}{2}(I - \hat{\sigma}_i^x)$$

$$i\hbar \frac{d}{dt}|\psi(t)\rangle = \widehat{H}(t)|\psi(t)\rangle$$

$$\widehat{H}(t) = \left(1 - \frac{t}{\tau}\right)\widehat{H}(0) + \left(\frac{t}{\tau}\right)\widehat{H}_{protein}$$

$$\widehat{H}_{protein} = \widehat{H}(t); \;\; t = \tau$$

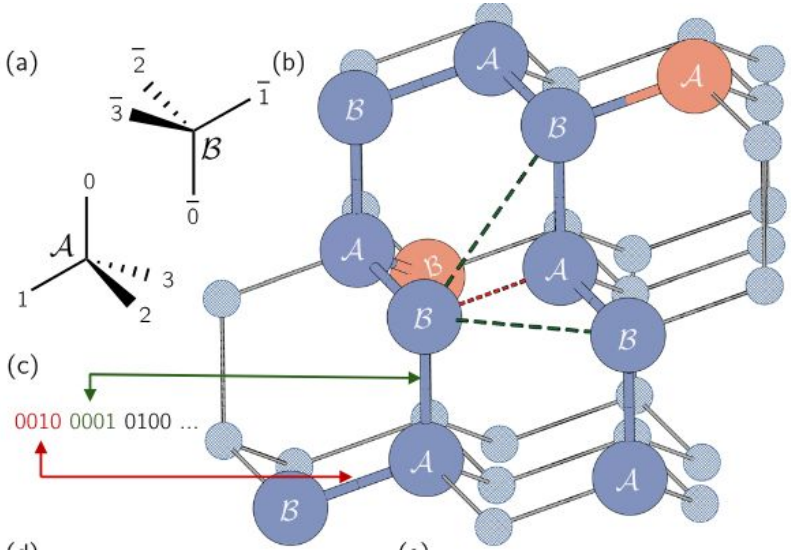
Hamiltonian Terms for Protein Folding

$$H_{protein} = H_{onsite} + H_{psc} + H_{pairwise}$$

- $\boldsymbol{H_{onsite}}$: Onsite repulsion term for amino acids occupying the same grid point.
- $\boldsymbol{H_{psc}}$: Primary sequence constraint term.
- $\boldsymbol{H_{pairwise}}$: Pairwise interaction term that represents favorable hydrophobic interactions between adjacent hydrophobic amino acids.

From node to edge encoding

- Tetrahedal lattice
- Encoding turns instead of coordinates
- One turn encoded via 4 qubits
- Geometric constraints are realized in Hamiltonian



Robert, A., Barkoutsos, P.K., Woerner, S. *et al.* Resource-efficient quantum algorithm for protein folding. *npj Quantum Inf* **7**, 38 (2021).

Fold your protein

A K I Q D K E G I P → H P H H H P P H P P → Random configuration (cf) on tetrahedal lattice

$$H_{protein} = H_{onsite} + H_{psc} + H_{pairwise}$$

Romero, S.V., Cadavid, A.G., Nikačević, P. et al. Protein folding with an all-to-all trapped-ion quantum computer. (2025).

Robert, A., Barkoutsos, P.K., Woerner, S. *et al.* Resource-efficient quantum algorithm for protein folding. *npj Quantum Inf* **7**, 38 (2021).

Conclusion

- **Mapping a complex problem to a simpler model (p spin glass) allows to use current available Quantum platforms**

- **Mapping decomposes the hardness of the problem, questioning the necessity of quantum resources and scaling the problem as a whole**

- **This solution gives us the basic structure to understand the behaviour of proteins.**

Thanks for your attention!

Cited papers

Robert, A., Barkoutsos, P.K., Woerner, S. *et al.* **Resource-efficient quantum algorithm for protein folding**. *npj Quantum Inf* **7**, 38 (**2021**). https://doi.org/10.1038/s41534-021-00368-4

Romero, S.V., Cadavid, A.G., Nikačević, P. et al. **Protein folding with an all-to-all trapped-ion quantum computer.** Preprint at https://arxiv.org/abs/2506.07548 (**2025**).

A. Lucas, "**Ising formulations of many NP problems**," *Front. Phys.*, vol. 2, Feb. **2014**, doi: 10.3389/fphy.2014.00005.

Perdomo-Ortiz, A., Dickson, N., Drew-Brook, M. *et al.* **Finding low-energy conformations of lattice protein models by quantum annealing.** *Sci Rep* **2**, 571 (**2012**). https://doi.org/10.1038/srep00571

Resource demands of the quantum simulations

2012 2021

	Perdomo *et al.* [1]	Perdomo *et al.* [2]	Babbush *et al.* [3]	Babej *et al.* [4, 5]	This model
Model	Hydrophobic-Polar (HP)	Coarse-Grained	Coarse-Grained	Coarse-Grained	Coarse-Grained
Lattice	all	all	all	all	Tetrahedral
Types	2	∞	∞	∞	∞
Interactions	Nearest	Nearest	Nearest	Nearest	l^{th} Nearest
Locality	$\log_2 N$	N	4	N	$l+2$
Qubits	$N\log(N)$	$N^2\log(N)$	$N^3\log(N)$	N	$N^2\exp(l)$
Scaling	N^8	$\exp(N)$	$N^{12}\log^4(N)$	$\exp(N)$	N^4
Experiment	No	D-Wave	No	Rigetti QPU	IBM QPU

Robert, A., Barkoutsos, P.K., Woerner, S. *et al.* Resource-efficient quantum algorithm for protein folding. *npj Quantum Inf* **7**, 38 (2021).